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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/776,642	07/18/2024	Richard A. Giambruno II	1009-0002	7765
185208	7590	07/16/2026	EXAMINER	
Kiklis and Clark, PLLC			SHUI, MING	
600 14th Street			ART UNIT	PAPER NUMBER
5th Floor			2663	
Washington, DC 20005			NOTIFICATION DATE	DELIVERY MODE
			07/16/2026	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary

Application No.

18/776,642

Applicant(s)

Giambruno, Richard A.

Examiner

Ming Shui

Art Unit

2663

AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1) ☒ Responsive to communication(s) filed on 6/18/26.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2a) ☒ This action is **FINAL**.

2b) ☐ This action is non-final.

3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☒ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) ____ is/are withdrawn from consideration.

6) ☐ Claim(s) ____ is/are allowed.

7) ☒ Claim(s) 1-20 is/are rejected.

8) ☐ Claim(s) ____ is/are objected to.

9) ☐ Claim(s) ____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10) ☐ The specification is objected to by the Examiner.

11) ☒ The drawing(s) filed on 7/18/24 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some** c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. ____.

3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

3) ☐ Interview Summary (PTO-413)

Paper No(s)/Mail Date ____.

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)

4) ☐ Other: ____.

Paper No(s)/Mail Date ____.

Notice of Pre-AIA or AIA Status

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

DETAILED ACTION

Response to Arguments

With respect argument 1, applicant argues against the references individually. In response to applicant's arguments against the references individually, one cannot show nonobviousness by attacking references individually where the rejections are based on combinations of references. See *In re Keller*, 642 F.2d 413, 208 USPQ 871 (CCPA 1981); *In re Merck & Co.*, 800 F.2d 1091, 231 USPQ 375 (Fed. Cir. 1986).

With respect to argument 2, the examiner apologizes for the typographical error of paragraph 24 when paragraph 24-26 was meant to be referenced. There is a notification output the heads up display that for a new route and that meets the elements of the claim. Applicant is also directed to paragraphs 53 and 66.

With respect to argument 3, applicant appears to concede that Lee's emergency vehicles include a police car and the use of images for detection. Therefore Lee discloses a law enforcement vehicle and image recognition. That Lee can detect other vehicles and use other sensors as part of its system does not change that Lee discloses applicant's claim elements.

With respect to argument 4, applicant is advised to read the entirety of Lee for full context of the statements in the cited paragraph. For example, Lee ¶¶24-28 describe the system that carries out the process described in ¶21. See also ¶¶79-88 describing Fig 4.

With respect to argument 5, there is no suggestion that the combination of Lee and Mobileye would result throwing out all of Lee's multi-modal classifier with only an image only machine learning model.

With respect to argument 6, motivation to combine is not based on whether the motivation is contrary to applicant's invention, only if one would combine with the cited references and there is clearly a motivation to combine Lee with Mobileye. Regardless, applicant's invention can be utilized in an vehicle as applicant's own specification indicates that this may be utilized in a user vehicle.

With respect to argument 7, Lee teaches an autonomous vehicle, Mobileye teaches parts of the autonomous vehicle not explicitly described in Lee. Therefore, the two references are properly combinable as one reference merely describes in greater detail aspects of the other.

With respect to argument 8, Lee ¶46 also notes that even in an autonomous vehicle, a driver may choose to drive the vehicle instead of relying solely on the autonomous driving. Regardless, Lee makes clear that knowing where law enforcement or other emergency vehicles are located is necessary so that the car can move out of the way of the emergency vehicle.

With respect to claim 9, applicant again refers to the purpose of the invention, but claims are not evaluated on their purpose, they are evaluated on the claim language itself and Lee meets the recited claim language. Further, the purpose of a claim appears to be an intended use, which is given little patentable weight.

With respect to argument 10, Lee is both in the same field of endeavor and reasonably pertinent to the inventor's problem. The invention is directed to systems in vehicles, specifically a system that can detect law enforcement vehicles, of which, Lee detects law enforcement vehicles and is therefore

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pertinent to the problem. Applicant's narrow framing of the problem ignores all the other problems that must be solved in order to solve applicant's purported problem.

With respect to dependent claim 4, in order to determine emergency vehicles that include police cars, it would necessarily have included training on police cars.

With respect to argument 1 of claim 6, applicant is free to amend "light device" to specify what they mean.

With respect to argument 2 of claim 6 again, applicant's invention is directed to vehicle systems, which Suh is also directed to.

With respect to argument 3 of claim 6, there is no requirement that the reasons for combining be the same as applicant's.

With respect to argument 4 of claim 6, applicant is free to amend "light device" to be specific to applicant's intent.

The examiner has therefore considered all of applicant's arguments and do not find them persuasive. The rejections are maintained.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

Claims 1-2, 4-5, 10-12, 14-15, and 20 are rejected under 35 USC 103 as being unpatentable over US 2019/0039613 Lee in view of How Autonomous Vehicles Work: the Self driving stack, Mobileye, May 31, 2023. (hereafter Mobileye)

1. A method in a data processing system for detecting a law enforcement vehicle, the method comprising,

using a processor of the data processing system: receiving a digital image, at the processor, from a camera in electronic communication with the processor, wherein the camera is disposed within a user vehicle; (Lee ¶21 sensor is a camera and is processed by a processor, thus there is a digital image and the apparatus is in the car see e.g. Fig 4A)

detecting the law enforcement vehicle within the digital image by the processor; and (Lee ¶21 uses the sensor to detect emergency vehicles, ¶21 discloses that emergency vehicles include police cars)

transmitting, by the processor, a notification instruction to a notification output device disposed within the user vehicle indicating the law enforcement vehicle was detected. (Lee ¶46 outputs information about the detection of an emergency vehicle)

Lee does not disclose

using a machine learning model, the machine learning model trained by training data including images of law enforcement vehicles

However, Mobileye discloses using a machine learning model that uses trained object recognition, pattern, recognition, etc. to recognize particular items in the image – to recognize emergency vehicles such as law enforcement vehicles, the algorithm would have been trained with images that included law

enforcement vehicles. It would have been obvious to modify the system of Lee to use machine learning because the use of machine learning is a fundamental part of creating autonomous vehicles.

Claim 11 is rejected under a similar rationale.

2. The method of claim 1, wherein the processor, the camera, and the notification output device are disposed within a housing. (Lee ¶69 smart phone, instrument cluster)

Claim 12 is rejected under a similar rationale.

4. The method of claim 1, further comprising training the machine learning model by the training data including images of law enforcement vehicles. (Mobileye discloses using a machine learning model that uses trained object recognition, pattern, recognition, etc. to recognize particular items in the image – to recognize emergency vehicles such as law enforcement vehicles, the algorithm would have been trained with images that included law enforcement vehicles.)

Claim 14 is rejected under a similar rationale.

5. The method of claim 1, wherein the processor, the camera, or the notification output device is integrated into the user vehicle. (Lee ¶68 included in vehicle)

Claim 15 is rejected under a similar rationale.

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10. The method of claim 1, wherein the notification output device includes a mobile device, and transmitting the notification instruction to the notification output device includes transmitting a communications signal to the notification output device via a networking device. (Lee ¶169 portable device through network)

Claim 20 is rejected under a similar rationale.

Claims 3 and 13 are rejected under 35 USC 103 as being unpatentable over Lee and Mobileye in view of US 2015/0122858, Azadi.

Lee and Mobileye do not disclose

3. The method of claim 2, wherein the housing includes a mounting device configured to secure the housing to a dashboard of the user vehicle.

However Azadi discloses a dash mount for devices. It would have been obvious to modify the system of Lee to include a dash mount for the purposes of providing a secure location for the portable device that does not interfere with the driver's handling of the vehicle or impair the vision of the driver as taught by Azadi.

Claim 13 is rejected under a similar rationale.

Claims 6-9 and 16-19 are rejected under 35 USC 103 as being unpatentable over Lee and Mobileye in view of US 2024/0174168 Suh.

6. The method of claim 1, wherein the notification output device is a light device, and the notification instruction includes a computer instruction to emit light from the light device. (Suh ¶24 provides a map on a screen, which is a light device)

It would have been obvious to modify the system of Lee and Mobileye to include a map for the purposes of providing real time information for the car as taught by Suh.

Claim 16 is rejected under a similar rationale.

7. The method of claim 1, wherein the notification output device includes a speaker, and the notification instruction includes a computer instruction to emit a sound from the speaker. (Suh Claim 15, voice message about approaching emergency vehicle)

It would have been obvious to make the modification for the same reasons as claim 16.

Claim 17 is rejected under a similar rationale.

8. The method of claim 7, wherein the sound is a generated voice statement indicating a location of the law enforcement vehicle relative to the user vehicle. (Suh Claim 15, voice message about approaching emergency vehicle)

It would have been obvious to make the modification for the same reasons as claim 16.

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Claim 18 is rejected under a similar rationale.

9. The method of claim 1, wherein the notification output device includes a display configured to display a map, and the notification instruction includes a computer instruction to display a location of the law enforcement vehicle on the map. (Suh Fig 4A illustrates the vehicle)

It would have been obvious to make the modification for the same reasons as claim 16.

Claim 19 is rejected under a similar rationale.

Conclusion

THIS ACTION IS MADE FINAL. Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Ming Shui whose telephone number is (303)297-4247. The examiner can normally be reached on 7-5 Pacific Time, M-Th.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Greg Morse can be reached on 571-272-3838. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Ming Shui/

Primary Examiner, Art Unit 2663